



U.S. Department
of Transportation

Research and
Special Programs
Administration

JAN 14 1998

400 Seventh Street, S.W.
Washington, D.C. 20590

DOT-E 11739

EXPIRATION DATE: December 31, 1999

(FOR RENEWAL, SEE 49 CFR 107.109)

1. **GRANTEE:** Oceaneering Space Systems
Houston, TX
2. **PURPOSE AND LIMITATIONS:**
 - a. This exemption authorizes the manufacture, mark, sale and use of a non-DOT specification breathing apparatus to be used for the transportation in commerce of air, refrigerated liquid. This exemption provides no relief from any Hazardous Material Regulation other than as specifically stated herein.
 - b. An exemption authorization to manufacture, mark, sell, and transport only represents certification of safety for a package when it is an article of commerce in transportation. The safety analyses performed in development of this exemption only considered the hazards and risks associated with transportation in commerce. The safety analyses did not consider the hazards and risks associated with consumer use, use as a component of a transport vehicle or other device, or other uses not associated with transportation in commerce.
3. **REGULATORY SYSTEM AFFECTED:** 49 CFR Parts 106, 107 and 171-180.
4. **REGULATIONS FROM WHICH EXEMPTED:** 49 CFR Sections 173.316(c) in that the prescribed packaging is not listed as an authorized packaging, and portions of 178.57 as specified herein.
5. **BASIS:** This exemption is based on the application of Oceaneering Space Systems (OSS) dated July 11, 1996 submitted in accordance with 49 CFR 107.105 and the public proceeding thereon.

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6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous materials description -- proper shipping name	Hazard Class/ Division	Identi- fication Number	Packing Group
Air, refrigerated liquid	2.2	UN1003	N/A

7. PACKAGING(S) and SAFETY CONTROL MEASURES:

a. PACKAGING - Packaging prescribed is a non-DOT specification breathing apparatus consisting of an injection molded plastic inner vessel within a steel or titanium vacuum jacket outer vessel. The inner vessel contains the air, refrigerated liquid and must be designed with a burst pressure of no less than 2.5 times the service pressure. The container must be in conformance with the DOT-4L cylinder (§178.57) except as follows:

§178.57-2 Type, Size, Service Pressure, and Design Service Temperature.

(a) Type and size. Inner vessel halves must be glued, and outer vessel halves must be fusion welded. Size must not be over 17 pounds water capacity.

(b) The service pressure must be 40 psig maximum.

(c) The design service temperature is -320°F.

§178.57-4 Duties of inspector.

(a) * * *

(b) Verify physical properties of each lot of raw material by analysis or by obtaining certified analysis: Provided that a certificate from the manufacturer thereof, giving sufficient data to indicate compliance with requirements, is acceptable when verified by check analyses of ASTM D638 Standard Slabs molded concurrently with the inner vessel.

(c) Verify compliance with all specification requirements. Obtain samples for all tests. Obtain samples for check physical properties, where required. * * *

(d) Furnish complete test reports required by this exemption to the maker of the inner vessel and, upon request, to the purchaser. * * *

§178.57-5 Material.

(a) Inner containment vessel. Designations and limiting physical properties of the liquid crystal polymer (LCP) authorized by this specification shall be as shown in §178.57-21(a) Table 1 of this exemption.

(b) Outer jacket. Titanium or steel may be used subject to the requirements of §178.57-21(b).

§178.57-7 Defects.

(a) Materials with seams, cold slugs, cracks, laminations, or other detrimental defects, not authorized. No defect acceptable that is likely to weaken the finished vessel appreciably; reasonably smooth and uniform surface finish required.

§178.57-8 Manufacture.

(a) Inner vessel. By injection molding of vessel halves and glueing; mold sprue and flashing, dirt and scale to be removed as necessary to afford proper inspection. Molding must be in accordance with the application for exemption. The vessel halves (heads) must be seamless. The heads must be reasonably true to shape, shall have no abrupt shape changes and the skirts must be reasonably true to round. The inner vessel halves and the center hub assembly shall be glued in accordance with OSS LCP Inner Vessel Epoxying Procedure dated 3/11/96 and modified 6/10/96.

(b) Outer vessel. The vessel halves (heads) must be seamless. By best appliances and methods; dirt and scale to be removed as necessary to afford proper inspection. When required, titanium shall be cleaned in accordance with ASTM B600 Standard Guide for Descaling and Cleaning Titanium and Titanium Alloy Surfaces.

(c) Insulation. The space between the inner vessel and the jacket shall be insulated. The insulating material must be fire resistant. If a vacuum is maintained in the insulation space, the evacuated jacket shall be designed for a minimum collapsing pressure of 30 psi differential whether made of steel or titanium. The construction must be such that the total heat transfer, from the atmosphere at ambient temperatures to the contents of the inner vessel, will not exceed 0.008 Btu per hour, per Fahrenheit degree differential in temperature, per pound of water capacity of the inner vessel.

§178.57-9 Welding.

(a) All seams of the outer jacket must be fusion welded. Only butt or joggle butt joints for the circumferential jacket seam are authorized. Transition penetrations or tube penetrations may be fillet welded. All joints in the outer vessel must be in reasonably true alignment.

(b) All attachments to the sidewalls and heads of the outer jacket must be by fusion welding and must be of a weldable material.

(c) For welding the outer jacket, each procedure and operator must be qualified in accordance with the sections of CGA Pamphlet C-3 that apply.

(d) Brazing and soldering are permitted only for joints not made directly to the outer jacket body.

§178.57-10 Wall thickness.

(a) Inner vessel. The minimum wall thickness of the inner containment vessel shall be calculated using Finite Element Analysis (FEA). Using FEA the calculated wall stress at minimum required test pressure shall not exceed one-third of the minimum tensile strength of the base material as required in §178.57-21.

(b) Outer vessel. The minimum wall thickness of the outer jacket shall be calculated using FEA. Using FEA the calculated wall stress at an external pressure of 30 psi shall not exceed the yield strength of the base metal determined as required in §178.57-21.

§178.57-11 Heat treatment.

(a) Inner vessel. Stress relief of the molded halves of inner vessel is permitted as necessary provided that the angle of the joining surfaces are within the drawing tolerances after heat treating (no distortions permitted).

(b) Outer vessel.

(1) Titanium. Titanium outer vessel may be vacuum annealed after chemical cleaning in accordance with ASTM B600 Standard Guide for Descaling and Cleaning Titanium and Titanium Alloy Surfaces.

(2) Steel. Heat treatment after forming is not permitted.

§178.57-12 Openings in Vessels.

(a) Inner vessel. Openings permitted in hub section only. The opening shall not exceed 0.19 square inches. Openings in the inner vessel shall access the center hub which shall be glued to the inner vessel. Attachments to the center hub may be made by welding, brazing, soldering, mechanical attachment, or threading.

(b) Outer vessel. Openings permitted only in areas not adjacent to knuckle radius. Center line of opening must be one diameter or more away from knuckle radius or weld seam. Openings must be circular and shall not exceed $\frac{1}{4}$ inch diameter. Attachments to the outer vessel may be made by welding, brazing, soldering, mechanical attachment, or threading.

§178.57-13 Pressure relief devices and pressure control valves.

Each finished assembly must be equipped with pressure relief devices and pressure control valves as prescribed in 49 CFR §173.34(d) and 173.316. Additionally the vessel shall be equipped with a rotating pick-up and vent assembly that maintains the vent in the ullage space regardless of vessel orientation.

§178.57-14 Pressure test.

(a) After assembly, each inner vessel, before insulating and jacketing must be examined under a pressure of at least two times the service pressure maintained for at least 30 seconds without evidence of leakage, visible distortion or other defect. Pressure gauge must permit reading to an accuracy of 1 percent.

§178.57-15 Physical test.

(a) Inner vessel. Concurrent with part molding, ASTM D638 Standard Slabs will be prepared from the same material and tested for physical properties to verify conformance with the minimum physical properties listed in §178.57-21(a) Table 1 of this exemption. Determine yield strength, tensile strength, and elongation on 2 specimens selected from each lot of raw material and in the same condition as that of the completed vessel.

(b) Glue joint. Following the OSS gluing procedure, LCP Inner Vessel Epoxying Procedure, prepare two specimens using ASTM D638 Standard Slabs. The slabs shall be continuously glued over their entire width and overlap each other one inch. The samples shall be cured and allowed to set for 7 days. Samples will then be pulled to failure in the tensile test apparatus.

§178.57-16 Acceptable results for physical tests.

(a) Physical properties must meet the limits specified in §178.57-21(a) Table 1 of this exemption for the LCP.

(b) Failure of the glued specimens must occur only in the base material.

§178.57-17 Tests of welds.

Not required.

§178.57-20 Marking.

(a) * * *

(1) DOT-E 11739 in lieu of DOT-4L, followed by the service pressure.

* * *

§178.57-21 Authorized materials of construction.

(a) Inner containment vessel. Liquid Crystal Polymer of uniform quality. Chemical analysis must conform to Hoechst Celanese Vectra A130. The following chemical analyses and physical properties are authorized:

TABLE 1 - AUTHORIZED MATERIALS

<u>Designation</u>	<u>Chemical Analysis, Limits in Percent</u>	
Glass Fiber	29 to 31	
Vectra Copolyester	Remainder	
Resin Grade A950		
<u>Physical Properties</u>		
Tensile strength at break, psi (minimum)		30,000
Tensile modulus, psi		3,200,000
Flexural strength, psi		37,000
Flexural modulus, psi		2,100,000
Compressive strength, psi (deflection)		20,000

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Compressive modulus, psi	1,700,000
Shear strength, psi	17,800
Izod impact strength, notched ft-lb/in	2.8
Tensile impact strength, ft-lb/in ²	40
Elongation at break, percent	2.2

CHECK ANALYSIS TOLERANCES

Elements	Limit or maximum of specified range, percent	Permissible variation in product analysis
Glass fiber	30	±1.0

(b) Outer vessel. Must be constructed of steel or titanium. Steel: Electric furnace steel of uniform quality. Chemical analysis must conform to ASTM A240, Type 304 stainless steel. Titanium: Alloy Ti-6Al-4V must conform to AMS 4911H.

b. OPERATIONAL CONTROLS - The service life of the prescribed container is limited to 15 years.

8. SPECIAL PROVISIONS:

a. Offerors for transportation of the hazardous materials specified in this exemption may use the packaging described in this exemption for the transportation of such hazardous materials provided no modifications or changes are made to the packages, all terms of this exemption are complied with, and a copy of the current exemption is maintained at each facility from which such offering occurs.

b. Each packaging manufactured under the authority of this exemption must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous Materials Exemptions and Approvals Program for a specific manufacturing facility.

c. A copy of this exemption, in its current status, must be maintained at each manufacturing facility at which this packaging is manufactured and must be made available to a DOT representative upon request.

d. Shippers using the packaging covered by this exemption must comply with all provisions of this exemption, and all other applicable requirements contained in 49 CFR Parts 171-180 that apply to the hazardous materials authorized and to a DOT Specification 4L cylinders.

- e. Design qualification testing must be performed in accordance with the OSS July 11, 1996 application for exemption, with acceptable results. The final test report must be on file with the Office of Hazardous Materials Exemptions and Approvals before initial shipment.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, and cargo aircraft only.
10. MODAL REQUIREMENTS: A copy of this exemption must be carried aboard each cargo vessel or aircraft used to transport packages covered by this exemption. The shipper shall furnish a copy of this exemption to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this exemption and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. Section 5101 et seq:
- o All terms and conditions prescribed in this exemption and the Hazardous Materials Regulations, Parts 171-180.
 - o Registration required by 49 CFR 107.601 et seq., when applicable.
- Each "Hazmat employee", as defined in 49 CFR 171.8, who performs a function subject to this exemption must receive training on the requirements and conditions of this exemption in addition to the training required by 49 CFR 172.700 through 172.704.
- No person may use or apply this exemption, including display of its number, when the exemption has expired or is otherwise no longer in effect.
12. REPORTING REQUIREMENTS: The carrier is required to report any incident involving loss of packaging contents or packaging failure to the Associate Administrator for Hazardous Materials Safety (AAHMS) as soon as practicable. (49 CFR 171.15 and 171.16 apply to any activity undertaken under the authority of this exemption.) In addition, the holder(s) of this exemption must also inform the AAHMS, in writing, as soon as practicable of any incidents involving the package and shipments made under this exemption.

Issued at Washington, D.C.

JAN 14 1998



Alan I. Roberts
Associate Administrator
for Hazardous Materials Safety

(DATE)

Address all inquiries to: Associate Administrator for Hazardous
Materials Safety, Research and Special Programs Administration,
Department of Transportation, Washington, D.C. 20590.
Attention: DHM-31.

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